

Simple IR Remote Control Extender

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Sometimes you wish you could extend the range of the remote control of your TV set or DVD player, or take it to another room for use. This circuit can receive the IR signal from

your remote control and retransmit it to the intended destination/equipment, even around a corner or to another room.

Circuit and working

Fig. 1 shows circuit diagram of the simple IR remote control extender. It is built around op-amp IC 741 (IC1), transistor CK100 (T1), photo-diode/receiving LED (IRRX1), transmitting LEDs (IRTX1 and IRTX2) and a few other components.

IRRX1 is basically a photo-diode, such as BPW41 or its equivalent BPW34. Since a photo-diode is designed to

signal in the form of a train of pulses is available at IC1's output pin 6. This signal goes to transistor T1 through current-limiting resistor R8. Transistor T1 drives the IR transmitters IRTX1 and IRTX2. Thus the IR signal received at IRRX1 is amplified and retransmitted to the equipment through the IR transmitters.

The current flowing through IRTX1 and IRTX2 should not exceed 100mA. Hence a fixed current-limiting resistor (R7) is used in series with preset VR1. LED1, an ordinary LED, flashes when an IR signal is received from the remote for retransmission.

To control TV or air-conditioner from a longer distance or another room, connect the photodiode (IRRX1) with a pair of long wires as shown in Fig. 2.

Construction and testing

An actual-size, single-side PCB for the simple infrared control extender is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable plastic case.

Fix a 2-pin connector (CON1) for power supply. Also fix a switch (S1) on the front panel for power on/off. Fix photo-diode IRRX1 in such a way that the remote's signal can easily fall on it. Keep the orientation of IR LEDs IRTX1 and IRTX2 such that the IR signals received at IRRX1 can be retransmitted like a repeater to a final destination through them.

To check the working of the circuit, point the remote towards IRRX1 and press any key of the remote. If LED1 flashes it means the IR signal is being received and getting retransmitted.

Now point IRTX1 and IRTX2 towards your equipment and again press any key of the remote. If you see LED1 flashing, you should be able to control your equipment through this circuit from a longer distance now.

Note: The directivity of the IR LEDs IRTX1 and IRTX2 and use of a reflector can increase the range further. ●

The author is a regular contributor to EFM



PARTS LIST

Semiconductors:

- IC1 - 741 op-amp
- LED1 - 5mm LED
- T1 - CK100 npn transistor
- IRTX1, IRTX2 - IR transmitting LED
- IRRX1 - IR receiving LED (BPW34 or BPW41)

Resistors (all 1/4-watt, ±5% carbon):

- R1, R3, R4 - 100-kilo-ohm
- R2 - 3.3-kilo-ohm
- R5 - 1-mega-ohm
- R6, R8 - 1-kilo-ohm
- R7 - 56-ohm
- VR1 - 470-ohm preset

Capacitors:

- C1 - 47nF ceramic disk

Miscellaneous:

- S1 - On/off switch
- CON1 - 2-pin terminal connector

Test Points

Test point	Details
TP0	0V
TP1	9V
TP2	Train of pulses when IR signal is received

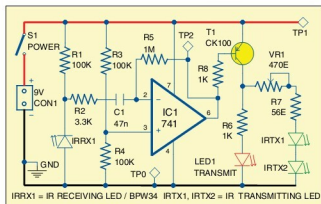


Fig. 1: Remote control extender's circuit diagram

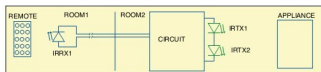


Fig. 2: Suggested wire extension of IR receiver between two rooms

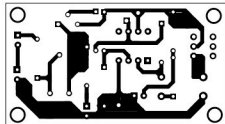


Fig. 3: Actual-size, single-side PCB layout for the circuit

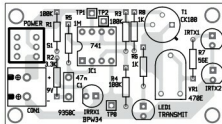


Fig. 4: Component layout for the PCB